

实验一：Hadoop Data Processing

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操作步骤

完成Java文件

Job1

1. 过滤并统一数据

```
public void map(Object key, Text value, Context context) throws IOException,
InterruptedException {
    StringTokenizer itr = new StringTokenizer(value.toString());
    while (itr.hasMoreTokens()) {
        //To Do job1: filter and uppercase the input data
        String raw = itr.nextToken();
        raw = raw.replaceAll("[^a-zA-Z0-9]", "");
        if (raw.isEmpty()) continue;           // 防止空串
        word.set(raw.toUpperCase());           // 统一大写
        context.write(word, one);
    }
}
```

2. 计数

```
public void reduce(Text key, Iterable<IntWritable> values, Context context)
throws IOException, InterruptedException {
    int sum = 0;
    //To Do job1: word count
    for (IntWritable val : values) {
        sum += val.get();
    }
    result.set(sum);
    context.write(key, result);
}

// TODO 2
sortJob.setSortComparatorClass(MyComparator.class);
```

Job2

直接调父类得到升序结果，再取反即为降序

```
private static class MyComparator extends IntWritable.Comparator {
    @Override
    public int compare(Object a, Object b) {
        //To Do job2: sort by descending order
        return -super.compare(a, b);
    }
    @Override
    public int compare(byte[] b1, int s1, int l1, byte[] b2, int s2, int l2)
    {
        //To Do job2: sort by descending order
        return -super.compare(b1, s1, l1, b2, s2, l2);
    }
}
```

验证

1. 编译

```
javac WordCountAndSort.java
```

2. 打包成jar包

```
jar -cvf WordCountAndSort.jar ./WordCountAndSort*.class
```

完成后目录如下

```
root@ubuntu:~/cloudz# ls
functions          output.txt          WordCountAndSort.jar
hadoop-2.7.3       python_3.7_slim_buster.tar  WordCountAndSort.java
hadoop-2.7.3.tar.gz WordCountAndSort.class      WordCountAndSort$MyComparator.class
openfaas           WordCountAndSort$IntSumReducer.class  WordCountAndSort$TokenizerMapper.class
root@ubuntu:~/cloudz# _
```

3. 检验hadoop服务是否启动，端口是否监听

- 输入命令 jps 检查是否启动

```
root@ubuntu:~/cloudz# jps
11058 NameNode
11682 ResourceManager
11490 SecondaryNameNode
11239 DataNode
13626 Jps
11998 NodeManager
root@ubuntu:~/cloudz# _
```

- 输入命令 netstat -nltp | grep 9000 查看端口监听状态

```
root@ubuntu:~/cloudz# netstat -nltp | grep 9000
tcp        0      0 127.0.0.1:9000        0.0.0.0:*              LISTEN      11058/java
root@ubuntu:~/cloudz#
```

4. 运行

```
nohup hadoop jar WordCountAndSort.jar WordCountAndSort /dataset1G
/sortResult > output.txt 2>&1 &
```

使用命令 tail output.txt -f 来查看输出

实验结果

```
root@ubuntu:~/cloudz# hadoop fs -cat /sortResult/part-r-00000 | head
100022  ARNEB
99873   PAUNCHY
99873   GROUNDNEEDLE
99862   ANGIOPATHY
99861   SAPONACEOUS
99842   AMBITUS
99806   LARYNGIC
99756   SIRRAH
99756   HACKNEYED
99752   APPROBATION
cat: Unable to write to output stream.
root@ubuntu:~/cloudz# A_
```

问题与解决

1. 9000端口未监听

输入命令 `jps` 时输出如果只有 `1306 Jps` 一行，说明其namenode没有启动，需要重新启动一下hadoop服务

```
/root/cloudz/hadoop-2.7.3/sbin/start-all.sh
```

在用 `jps` 检查一下，并且使用 `netstat` 命令检查端口是否监听

2. Input path does not exist

```
25/12/23 19:51:48 INFO mapreduce.JobSubmitter: Cleaning up the staging area file:/usr/local/hadoop/tmp/mapred/staging/root422564057/.staging/job_local422564057_0002
Exception in thread "main" org.apache.hadoop.mapreduce.lib.input.InvalidInputException: Input path does not exist: hdfs://127.0.0.1:9000/user/root/wordcount-temp-20712802
* at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:323)
  at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.listStatus(FileInputFormat.java:265)
  at org.apache.hadoop.mapreduce.lib.input.SequenceFileInputFormat.listStatus(SequenceFileInputFormat.java:59)
  at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.get_splits(FileInputFormat.java:387)
  at org.apache.hadoop.mapreduce.JobSubmitter.writeNewSplits(JobSubmitter.java:301)
  at org.apache.hadoop.mapreduce.JobSubmitter.writeSplits(JobSubmitter.java:318)
  at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:196)
  at org.apache.hadoop.mapreduce.Job$10.run(Job.java:1290)
  at org.apache.hadoop.mapreduce.Job$10.run(Job.java:1287)
  at java.security.AccessController.doPrivileged(Native Method)
  at javax.security.auth.Subject.doAs(Subject.java:422)
  at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1698)
  at org.apache.hadoop.mapreduce.Job.submit(Job.java:1287)
  at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1308)
  at WordCountAndSort.main(WordCountAndSort.java:147)
  at sun.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
  at sun.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
  at sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
  at java.lang.reflect.Method.invoke(Method.java:498)
  at org.apache.hadoop.util.RunJar.run(RunJar.java:221)
  at org.apache.hadoop.util.RunJar.main(RunJar.java:136)
```

解决方法：重新跑一次即可。该目录是运行时随机生成的。